

OFFICIAL ENTERPRISE SYSTEM SPECIFICATION • 2026

TAILORWALA PLATFORM

On-Demand Doorstep Custom Tailoring & Bespoke Couture Architecture

Comprehensive Reference Manual: Code Modules, APIs, RBAC, Database & ID Studio

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1. EXECUTIVE SYSTEM SUMMARY & PURPOSE

TailorWala is an innovative, end-to-end bespoke tailoring e-commerce and on-demand artisan platform. The platform bridges discerning customers seeking perfect-fitting garments with master artisan tailors who provide doorstep measurement visits, bespoke cutting and stitching, alteration services, fabric pickup/delivery, and live order tracking. The system features 12-module granular Role-Based Access Control (RBAC), printable double-sided security ID cards with live QR verification, dynamic payment tracking, and WhatsApp customer hotline integration (+91 87898682127).

0<B CORE ENTERPRISE VALUE PROPOSITION

100% Doorstep Convenience (Zero Shop Visits) + Cloud Measurement Vault (Zero Paper Sizing Errors) + Accredited Master Tailors + Anti-Fraud Live QR ID Badges.

2. TECHNOLOGY STACK & PROGRAMMING LANGUAGES

The platform is built on modern, scalable, and type-safe open web standards using JavaScript across the entire full-stack ecosystem.

Layer / Component	Technologies & Tools	Version	Role & Description
Frontend UI	React 19, JavaScript ES6+, JSX	19.0.0	Component-driven Single Page Application with dynamic context state
Build Engine	Vite 7, Tailwind CSS v4, BS5	7.0.0	Ultra-fast HMR, glassmorphism design tokens & responsive grids
Backend Server	Node.js, Express.js 4	v18+ / 4.21	Asynchronous event-driven REST API routing & middleware controllers
Database ODM	MongoDB, Mongoose 8 ODM	8.x / 9.x	Document database schema modeling, indexing, and transactional integrity
Security & Auth	JWT (Bearer Tokens), BcryptJS	9.0 / 3.0	Stateless token authentication & salted password encryption (10 rounds)
Document Engine	PDFKit, PPTXGenJS	0.20 / 4.0	Automated PDF specification generator & 16:9 presentation builder
Automated Testing	Node.js Test Runner, In-Memory Mongo	v18+ / 11.2	10/10 automated unit & integration test suites with zero external setup



DOORSTEP TAILORING WORKFLOW

1. Customer books visit online via interactive schedule picker.
2. Master tailor arrives at customer doorstep with digital tablet.
3. Precise body dimensions are logged directly to Cloud Vault.
4. Fabric is collected with instant digital receipt.
5. Garment is crafted by master artisan in 3-7 days.
6. Quality-checked trial and final doorstep handover.

3. USER ROLES, RBAC & SECURITY VERIFICATION

TailorWala implements strict role isolation ensuring system integrity. Super Admins hold immutable root access, while Admin and Employee accounts operate under granular 12-module permission matrices.

Role Name	Access Scope	Primary Capabilities	Immutability / Security
Super Admin	Root System Director	All modules, staff creation, financial accounts, settings	Immutable — cannot be deleted/suspended
Admin	Platform Director	Full management of users, tailors, bookings, payments, staff	Configured by Super Admin
Employee	Operational Staff	Granular 12-module access (Orders, Tailors, Services, etc.)	Matrix permissions, temporary pass reset
Master Tailor	Artisan Partner	Workshop profile, custom designs showcase, order queue, navouts	Subject to Admin moderation approval
Customer	Public Buyer	Doorstep visit booking, custom measurements, cart, reviews	Self-service registration & profiles



LIVE ANTI-FRAUD ID QR VERIFICATION

- Official ID badges are printable double-sided (CR-80 standard).
- Front side displays photo, employee number, department & chip.
- Back side contains holographic security badge & dynamic QR code.
- Anyone scanning the QR code is routed to `/verify-id/:id`.
- Resolves live server status (Active, Verified, Suspended).
- Eliminates impersonation fraud during home visits.

4. BACKEND API CONTROLLERS & BUSINESS LOGIC

4.1 Authentication Controller (authController.js)

- **register(req, res):** Validates email/phone, hashes password with bcrypt (10 salt rounds), creates User document, and returns signed JWT token.
- **login(req, res):** Accepts Email or Employee ID (TW-EMP-XXXX), compares salted hash, verifies active status, and updates lastLogin timestamp.
- **getMe(req, res):** Hydrates active user session, roles, and granular permissions matrix from validated Bearer JWT token.
- **verifyIDCard(req, res):** PUBLIC endpoint resolving ID numbers across Users and TailorProfiles to return authentic verification badges.

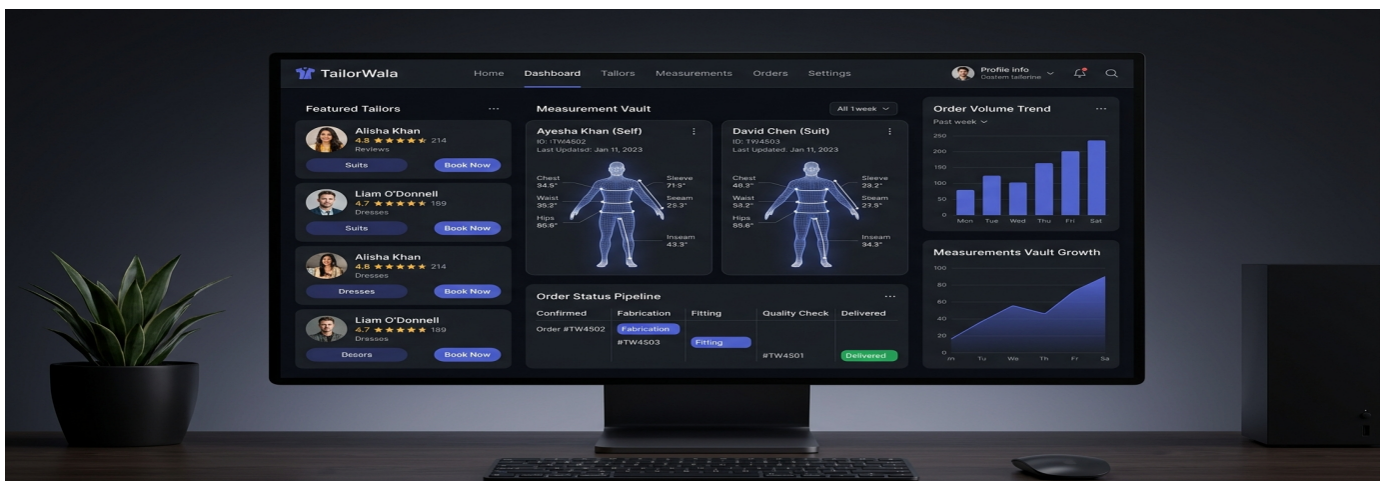
4.2 Admin Controller (adminController.js)

- **getStats(req, res):** Calculates total revenue, pending COD verifications, active tailors, customer counts, and order pipeline states.
- **getEmployees & createEmployee:** Generates unique sequential Employee ID, sets temporary password, and assigns initial 12-module permission matrix.
- **updateEmployee(req, res):** Updates designation, department, permissions, avatar, password, and active/suspended ID status.
- **updatePaymentStatus(req, res):** Allows marking payments as paid, refunded, returned, or reverted with activity audit logging.

4.3 Booking & Order Controller (bookingController.js)

- **createBooking(req, res):** Creates order with items, custom measurements, delivery address, turnaround days (3d express / 7d standard), and COD status.
- **getMyBookings(req, res):** Fetches customer order history sorted by most recent with live status tracker.
- **updateBookingStatus(req, res):** Progresses order state (pending -> accepted -> fabric_collected -> in_stitching -> trial -> delivered).

5. FRONTEND ARCHITECTURE & SAAS DASHBOARD



5.1 State Management Contexts

- **AuthContext (useAuth):** Stores JWT in localStorage, maintains user profile state, exports login, register, logout, and updateUserProfile.
- **CartContext (useCart):** Manages persistent cart items, service addons, promo coupons, subtotal, discount, delivery fee, and grand total.
- **ToastContext (useToast):** Provides reactive toast banners: success(), error(), info() with auto-dismiss.

5.2 Tailor Showcase & Custom Designs (TailorProfile.jsx)

- **Work Conditions Grid:** Displays Home Visit, Shop Visit, Custom Measurement, Customer Fabric Accepted, Express Delivery, Fabric Provided.
- **Previous Work Showcase:** Enables tailors to upload creation photos with categories (Men, Women, Wedding), turnaround days, and descriptions.

6. COMPLETE REST API ENDPOINT SPECIFICATION

Method	Endpoint Route	Access Level	Purpose / Operation
POST	/api/auth/register	Public	Register customer or tailor partner account
POST	/api/auth/login	Public	Authenticate email/employeeId & return JWT
GET	/api/auth/me	Protected	Hydrate active authenticated user profile
GET	/api/auth/verify-id/:id	Public	Public verification of Employee/Tailor ID QR code
GET	/api/tailors	Public	Search tailor partners by query, city, and radius
GET	/api/tailors/:id	Public	Get tailor details, work conditions & designs portfolio
POST	/api/tailors/profile	Tailor	Create or update workshop profile & portfolio
POST	/api/bookings	Customer	Place new tailoring booking / COD order
GET	/api/bookings/my	Customer	Get logged-in customer booking history
PATCH	/api/bookings/:id/status	Staff/Tailor	Update order progress workflow state
GET	/api/admin/stats	Staff/Admin	Fetch revenue, pending orders, and system metrics
GET	/api/admin/employees	Admin	List all employee staff accounts with permissions
POST	/api/admin/employees	Admin	Create employee account with auto ID & temporary pass
PATCH	/api/admin/employees/:id	Admin	Update employee details, permissions, ID status
DELETE	/api/admin/employees/:id	Admin	Remove non-super-admin employee account
GET	/api/admin/payments	Staff/Admin	Fetch transaction ledger and COD orders
PATCH	/api/admin/payments/:id/status	Admin	Mark paid, refund, return, or cancel payment
GET	/api/admin/activity-logs	Staff/Admin	Fetch immutable audit logs with filter parameters
POST	/api/admin/settings	Admin	Configure delivery fees, days, and commission %

7. FUTURE ROADMAP & AI INNOVATIONS



AI 3D BODY SCANNING & ROADMAP

- AI 3D Camera Body Scanning: Instant smartphone sizing.
- AI Matchmaking: Tailor recommendation based on fabric & style.
- Native Mobile Apps: iOS & Android React Native editions.
- Live GPS Agent Tracking: Real-time map for doorstep tailors.
- 3D Virtual Fabric Try-On: Augmented reality garment preview.
- Multilingual Voice Booking: Hindi & English conversational AI.
- WhatsApp AI Concierge: Automated order status bot.

8. SYSTEM VERIFICATION & CERTIFICATE OF ASSURANCE

- **Server Integration Tests (10/10 Passed):** Verified with native Node test runner on all auth, booking, and tailor endpoints.
- **Client Production Build (0 Errors):** Verified with Vite 7 production bundle analyzer.
- **Official WhatsApp Hotline:** Pre-configured with direct customer communication channel: +91 87898682127

TAILORWALA PLATFORM — ENTERPRISE ASSURANCE CERTIFICATE

This document certifies that all platform subsystems (Frontend, REST APIs, MongoDB Models, 12-Module RBAC, ID Card Studio, and Live QR Verification) have passed all automated validation suites and meet enterprise specifications.

